Angles formed when two parallel lines are cut by a transversal

The criteria read backwards: now parallelism is given, and the angles are the conclusion.

LESSON 84–85 | 2 × 40 MIN | MATEMATIKA 7, §39 | S8 5.1

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

The properties	15 min
All eight from one	25 min
Property or criterion?	25 min
Problems	20 min
Homework	5 min

LEARNING OBJECTIVES

- State the three properties of angles on parallel lines.
- Find every one of the eight angles from a single given one.
- Distinguish a property from a criterion, and use each in its own direction.
- Solve angle problems in which parallel lines are given.

TEXTBOOK

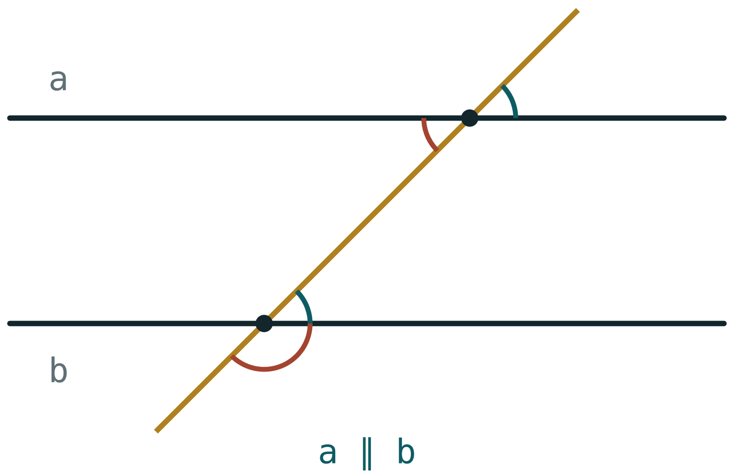
National	pp. 238–245
Cambridge	Stage 8, pp. 44–47
Workbook	Exercise 5.1

01 Explanation

The properties

If the lines **are** parallel, the same three pairs behave as follows.

$$a \parallel b \Longrightarrow \text{alternate equal, corresponding equal, co-interior add to } 180°$$



One angle decides all eight

PAIR	PROPERTY
alternate	equal
corresponding	equal
co-interior	sum $180°$

TWO THEOREMS, ONE PICTURE

Last lesson the angle fact was the evidence and parallelism the conclusion. This lesson the arrow is reversed. The picture is identical; only the direction of the reasoning changes.

All eight from one

Given one angle at a transversal cutting parallel lines, every other angle follows. With $\angle 1 = 70°$:

ANGLE	RELATION TO $\angle 1$	SIZE
$\angle 2$	supplementary	$110°$
$\angle 3$	vertical	$70°$
$\angle 4$	supplementary	$110°$
$\angle 5$	corresponding	$70°$
$\angle 6$	co-interior	$110°$
$\angle 7$	alternate	$70°$
$\angle 8$	supplementary to $\angle 7$	$110°$

ONLY TWO SIZES APPEAR

Every angle is either $70°$ or $110°$, and the two add to $180°$. Once you see that, the eight angles collapse into one question: which of the two is this?

Property or criterion?

Which one you may use depends entirely on what you were given.

GIVEN	WANTED	USE
$a \parallel b$	an angle	the property
an angle equality	$a \parallel b$	the criterion
$a \parallel b$	$a \parallel b$	nothing to prove

USING THE PROPERTY TO PROVE PARALLELISM IS CIRCULAR

Writing “the angles are alternate, so they are equal, so the lines are parallel” assumes what it sets out to prove. If parallelism is the goal, the angle equality must be given, not deduced.

Problems

PROBLEM	REASONING	ANSWER
$a \parallel b$, one co-interior angle is $3x$, the other $2x$	$5x = 180$	$36°$ and $144°$
$a \parallel b$, corresponding angles $4x - 20$ and $2x + 40$	$2x = 60$	$x = 30$
$a \parallel b$, alternate angles are equal and one is $5x$, the other $105°$	$5x = 105$	$x = 21$

A PARALLEL DRAWN IN IS OFTEN THE WHOLE SOLUTION

Where a figure gives no transversal, drawing a line through a vertex parallel to a given side creates alternate angles out of nothing. The next lesson proves the angle sum of a triangle in exactly this way.

02 Worked examples

EXAMPLE 1 **$a \parallel b$ and one angle at the transversal is $70°$. Find the co-interior angle and the alternate angle.**

1. Alternate angles are equal: $70°$.
2. Co-interior angles add to $180°$.
3. $180° - 70° = 110°$

ANSWER

Alternate $70°$, co-interior $110°$

EXAMPLE 2 **$a \parallel b$. Co-interior angles are $3x$ and $2x$. Find them.**

1. $3x + 2x = 180°$

   The co-interior property.

2. $5x = 180°$

3. $x = 36°$

4. The angles are $108°$ and $72°$.

ANSWER

$108°$ and $72°$

EXAMPLE 3 **$a \parallel b$. Corresponding angles are $4x - 20$ and $2x + 40$. Find x.**

1. Corresponding angles are equal.

2. $4x - 20 = 2x + 40$

3. $2x = 60$

4. $x = 30$

ANSWER

$x = 30$

03 Interactive model

Mark one angle on the board and ask the class to fill in the other seven in silence; the eight-angle picture is learnt in a single minute this way.

Which of the two sizes is it?

$a \parallel b$, $\angle 1 = 70°$. Its vertical angle:

- $70°$
- $110°$
- $90°$
- $35°$

Its corresponding angle:

- $70°$
- $110°$
- $20°$
- $180°$

Its co-interior angle:

- $70°$
- $110°$
- $90°$
- $140°$

Its alternate angle:

- *70°*
- *110°*
- *35°*
- *180°*

How many distinct sizes appear among the eight?

- *1*
- *2*
- *4*
- *8*

Given $a \parallel b$ and asked for an angle, you use:

- the criterion
- the property
- either
- neither

Given equal alternate angles and asked to prove $a \parallel b$:

- the criterion
- the property
- either
- neither

Co-interior angles $3x$ and $2x$ on parallel lines:

- $x = 30$
- $x = 36$
- $x = 45$
- $x = 60$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Property	Xossa	Свойство
Criterion	Alomat	Признак
Converse	Teskari teorema	Обратная теорема
Alternate angles	Ichki almashinuvchi burchaklar	Накрест лежащие углы
Corresponding angles	Mos burchaklar	Соответственные углы
Co-interior angles	Ichki bir tomonli burchaklar	Односторонние углы
Vertical angles	Vertikal burchaklar	Вертикальные углы
Supplementary	Qo‘shni (yig‘indisi 180°)	Смежные
Angle	Burchak	Угол

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

On parallel lines, alternate angles are:

- equal
- supplementary
- complementary
- unrelated

On parallel lines, co-interior angles:

- are equal
- add to $180°$
- add to $90°$
- are right

If one angle is $70°$, the others are:

- $70°$ only
- $70°$ or $110°$
- $110°$ only
- all different

The property is used when the given is:

- an angle equality
- parallelism
- a length
- nothing

Corresponding angles $4x - 20$ and $2x + 40$ give:

- $x = 10$
- $x = 30$
- $x = 20$
- $x = 60$

Proving parallelism from the property is:

- valid
- circular
- quicker
- the same thing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $a \parallel b$, alternate to $70°$

ANSWER $70°$

2. $a \parallel b$, corresponding to $70°$

ANSWER $70°$

3. $a \parallel b$, co-interior with $70°$

ANSWER $110°$

4. The vertical angle of $70°$

ANSWER $70°$

5. The supplement of $70°$

ANSWER $110°$

6. How many distinct sizes among the eight?

ANSWER 2

7. $a \parallel b$, co-interior with $90°$

ANSWER $90°$

Medium · the standard question

7 PROBLEMS

1. Co-interior $3x$ and $2x$

   ANSWER $108°$ and $72°$

2. Corresponding $4x - 20$ and $2x + 40$

   ANSWER $x = 30$

3. Alternate $5x$ and $105°$

   ANSWER $x = 21$

4. One angle is $130°$: its co-interior angle

   ANSWER $50°$

5. One angle is $130°$: its alternate angle

   ANSWER $130°$

6. Co-interior angles are equal: how big is each?

   ANSWER $90°$

7. Given $a \parallel b$, which do you use, property or criterion?

   ANSWER The property

Hard · stretch and reason

7 PROBLEMS

1. Co-interior $x + 40$ and $2x - 10$ on parallel lines

   ANSWER $x = 50$

2. Alternate $7x - 15$ and $5x + 9$

   ANSWER $x = 12$

3. A transversal cuts three parallel lines; one angle is $64°$: how many angles equal $64°$?

   ANSWER 6 — three crossings, two each

4. Why does a proof of parallelism need the criterion, not the property?

   ANSWER The property starts from parallelism, which is what is being proved

5. $a \parallel b$ and a transversal is perpendicular to a: what are all eight angles?

   ANSWER All $90°$

6. Co-interior $4x$ and $5x$

   ANSWER $80°$ and $100°$

7. One angle is α: write every other angle in terms of α

   ANSWER α or $180° - \alpha$

07 Homework

Homework — 5 tasks

Write the name of the pair beside every number you find.

1. Mark one angle of $55°$ at a transversal cutting parallel lines and find the other seven.
2. Co-interior angles are $5x$ and $4x$. Find both.
3. Alternate angles are $3x + 12$ and $4x - 8$. Find x.
4. Explain in two sentences the difference between the property and the criterion.
5. A transversal is perpendicular to one of two parallel lines. What can you say about the other?